FORWARD DEPLOYED / BACKEND SYSTEMS

INVARIANT OPS

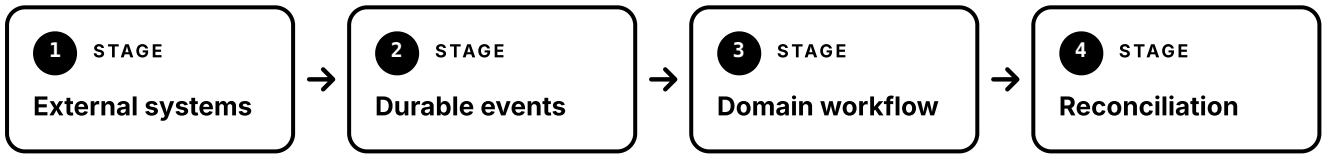
PROJECT 05

Correct enterprise workflows under ugly real world failure.

A failure prone procurement workflow built to demonstrate business correctness through retries, crashes, stale schemas, duplicate messages and partial transactions.

PROPOSED RESEARCH / NOT YET VERIFIED

SYSTEM / RESEARCH ARCHITECTURE



THE PROBLEM TO SOLVE

Which transaction and delivery patterns preserve business invariants when external integrations are unreliable?

TECHNICAL DISTINCTION

1. The key is preserving business truth, not showcasing infrastructure complexity for its own sake.
2. This is highly relevant for forward deployed and platform engineering if it is tied to explicit invariants.

WHAT THE EVIDENCE MUST SHOW

- No duplicate payment/posting under retry and replay
- State reconciliation after crash and restart
- Recovery latency and backlog under injected failure